End-to-End Data Management Pipeline for a Movie Recommendation System

BITS ID: 2025AG05520

1. Objective and design

This project implements the Data Management for Machine Learning Assignment 2 using the MovieLens Latest Small dataset. The goal is a repeatable workflow that validates and prepares interaction data, creates reusable features, trains two recommendation approaches, and records comparable offline metrics.

GitHub repository: recommendation-pipeline: https://github.com/Tango9100/recommendation-pipeline

The implementation uses modular Python scripts and a fail-fast root orchestrator. Project paths and model/evaluation settings are centralized in config/config.yml. CSV is used for transparent intermediate artifacts, and MLflow uses a local SQLite tracking store.

2. Data and processing

The input consists of 9,742 movie records and 100,836 user ratings from 610 users. MovieLens provides user identifiers but no demographic profile fields; therefore the user dimension table records IDs only and user activity features are derived from training interactions.

The validation stage checks required columns, missing values, duplicate rows, numeric/integer field types, rating bounds (0.5–5.0), convertible timestamps, and rating-to-movie references. Invalid rows are saved in data/errors/; the current dataset produced zero invalid type, rating, timestamp, or reference records and no missing or duplicate values.

Preparation standardizes identifiers and ratings, removes invalid records, and separates each user’s latest interaction into test data. This yields 100,226 training interactions and 610 held-out interactions. Transformations include user activity summaries, item popularity, and a user-item matrix.

3. Feature management and models

Feature engineering creates user features (average rating, activity count, unique movies, recency), item features (popularity, unique users, average rating, title and genres), and interaction features (rating, timestamp, hour, day of week, and month). Features used for training are generated from the training split to avoid leakage from held-out interactions. Their definitions are listed in reports/feature_catalog.csv.

The baseline popularity recommender ranks movies by training rating count and average rating, excluding items with fewer than five training ratings. The personalized recommender uses item-based collaborative filtering with cosine similarity and user rating history. Both models persist artifacts under models/.

4. Evaluation and results

The models are compared at K=5 on all 610 users with held-out interactions. Metrics are Precision@5, Recall@5, NDCG@5, catalog Coverage, and genre-based Diversity.

Model | Precision@5 | Recall@5 | NDCG@5 | Coverage | Diversity

Popularity | 0.0052 | 0.0262 | 0.0145 | 0.0077 | 0.7790

Collaborative Filtering | 0.0062 | 0.0311 | 0.0186 | 0.0612 | 0.7373

Collaborative filtering is higher on precision, recall, NDCG, and coverage for this run; popularity has higher diversity. The absolute ranking metrics are low, so the results should be treated as a reproducible baseline rather than evidence of production-quality recommendations. These recommenders rank items and do not predict ratings, so RMSE and MAE are not applicable.

5. Orchestration, tracking, and limitations

Run python run_pipeline.py from the repository root to execute ingestion, validation, preparation, transformation, feature engineering, both training stages, evaluation, and MLflow tracking in dependency order. A failure stops subsequent stages. MLflow records dataset and split parameters, K, ranking metrics, and model/report artifacts in the configured SQLite store.

The main limitations are the absence of user demographic attributes, reliance on one public dataset, and weak top-five accuracy. The models are offline recommenders, not a deployed service; feature storage is CSV-based rather than a production feature store. A future iteration should evaluate alternative splits and ranking algorithms and add temporal/user preference features from an appropriate data source.

PIPELINE EXECUTION EVIDENCE

Run completed: 2026-11-14 12:11:42 India Standard Time

Execution step screenshots: 9

The following pages show the captured terminal output and result for each executed step.

STEP 1: 1. Data Ingestion

SUCCESS | Execution time: 3.23 seconds

```
... 1 earlier output lines omitted ...
Data ingestion completed successfully!

Movies dataset:
Number of rows: 9742
Columns: ['movieId', 'title', 'genres']

Ratings dataset:
Number of rows: 100836
Columns: ['userId', 'movieId', 'rating', 'timestamp']

Sample movie records:
   movieId  ...                                       genres
0        1  ...  Adventure|Animation|Children|Comedy|Fantasy
1        2  ...                   Adventure|Children|Fantasy
2        3  ...                               Comedy|Romance
3        4  ...                         Comedy|Drama|Romance
4        5  ...                                       Comedy

[5 rows x 3 columns]

Sample rating records:
   userId  movieId  rating  timestamp
0       1        1     4.0  964982703
1       1        3     4.0  964981247
2       1        6     4.0  964982224
3       1       47     5.0  964983815
4       1       50     5.0  964982931
SUCCESS: 1. Data Ingestion
Execution time: 3.23 seconds
```

STEP 2: 2. Data Validation

SUCCESS | Execution time: 2.61 seconds

```
Starting data validation...

DATA VALIDATION RESULTS
=======================================
Missing values in movies:
movieId    0
title      0
genres     0
Missing values in ratings:
userId       0
movieId      0
rating       0
timestamp    0
Duplicate rows in movies: 0
Duplicate rows in ratings: 0
Movies columns validation: True
Ratings columns validation: True
Invalid movie field types: 0
Invalid rating field types: 0
Invalid ratings (outside 0.5-5.0 range or non-numeric): 0
Invalid movie references in ratings: 0
Invalid timestamps in ratings: 0

Data validation completed successfully!
Validation report saved to: C:\Users\Tanmay\Desktop\Recommendation-pipeline\data\errors\validation_report.txt
SUCCESS: 2. Data Validation
Execution time: 2.61 seconds
```

STEP 3: 3. Data Preparation

SUCCESS | Execution time: 4.66 seconds

```
... 7 earlier output lines omitted ...
Duplicate removal:
Movies duplicates removed : 0
Ratings duplicates removed: 0

============================================================
PREPARATION RESULTS
============================================================

Clean movies           : 9,742
Clean interactions     : 100,836
Training interactions  : 100,226
Testing interactions   : 610
Unique users           : 610
Unique movies rated    : 9,724
Minimum rating         : 0.5
Maximum rating         : 5.0
First interaction      : 1996-03-29 18:36:55
Last interaction       : 2018-09-24 14:27:30

Files created:
data/processed/movies_clean.csv
data/processed/ratings_clean.csv
data/processed/train.csv
data/processed/test.csv
data/processed/users.csv

Data preparation completed successfully.
SUCCESS: 3. Data Preparation
Execution time: 4.66 seconds
```

STEP 4: 4. Data Transformation

SUCCESS | Execution time: 14.65 seconds

```
... 8 earlier output lines omitted ...
3       4               216           216          3.556 2001-12-05 17:49:02
4       5                44            44          3.636 1996-11-08 06:42:17

Most popular movies:
      movieId  ... average_rating
314       356  ...          4.164
277       318  ...          4.429
257       296  ...          4.197
510       593  ...          4.161
1938     2571  ...          4.192
224       260  ...          4.231
418       480  ...          3.750
97        110  ...          4.032
507       589  ...          3.971
461       527  ...          4.225

[10 rows x 4 columns]

User-item matrix shape:
(610, 9724)

Files created:
data/features/user_activity.csv
data/features/item_popularity.csv
data/features/user_item_matrix.csv

Transformation completed successfully.
SUCCESS: 4. Data Transformation
Execution time: 14.65 seconds
```

STEP 5: 5. Feature Engineering

SUCCESS | Execution time: 4.30 seconds

```
... 23 earlier output lines omitted ...
Feature Catalog:
                      feature  ...                                        description
0                      userId  ...  MovieLens user identifier; no profile attribut...
1              user_avg_rating  ...                   Average rating given by the user
2          user_activity_count  ...                Total number of user interactions
3           user_unique_movies  ...             Number of unique movies rated by user
4   days_since_last_interaction  ...         Days since user's most recent interaction
5                      movieId  ...                           MovieLens item identifier
6              item_popularity  ...                  Total number of ratings received
7            item_unique_users  ...           Number of unique users rating the movie
8              item_avg_rating  ...                  Average rating received by movie
9                       genres  ...                           Movie genre information
10                      rating  ...    Explicit rating supplied by a user for a movie
11           interaction_time  ...       UTC timestamp converted to a datetime value
12           interaction_hour  ...         Hour of day when the interaction occurred
13     interaction_day_of_week  ...  Day of week when the interaction occurred, Mon...
14           interaction_month  ...                Month when the interaction occurred

[15 rows x 4 columns]

Files created:
data/features/user_features.csv
data/features/item_features.csv
data/features/interaction_features.csv
reports/feature_catalog.csv

Feature engineering completed successfully.
SUCCESS: 5. Feature Engineering
Execution time: 4.30 seconds
```

STEP 6: 6. Popularity Model Training

SUCCESS | Execution time: 2.80 seconds

```
... 6 earlier output lines omitted ...

 movieId                                    title  rating_count  average_rating
     356                      Forrest Gump (1994)           329           4.164
     318          Shawshank Redemption, The (1994)           314           4.430
     296                     Pulp Fiction (1994)           305           4.192
    2571                      Matrix, The (1999)           278           4.192
     593          Silence of the Lambs, The (1991)           278           4.162
     260 Star Wars: Episode IV - A New Hope (1977)           251           4.231
     480                    Jurassic Park (1993)           236           3.748
     110                       Braveheart (1995)           235           4.032
     589         Terminator 2: Judgment Day (1991)           221           3.959
    2959                      Fight Club (1999)           218           4.273

Movies eligible for recommendation: 3,633

Model saved to: C:\Users\Tanmay\Desktop\Recommendation-pipeline\models\popularity_model.pkl

Popularity model training completed successfully.

Top 5 recommendations for User 1:

 movieId                                               title  rating_count  average_rating
     318                       Shawshank Redemption, The (1994)           314           4.430
     589                      Terminator 2: Judgment Day (1991)           221           3.959
     150                                       Apollo 13 (1995)           200           3.840
    4993 Lord of the Rings: The Fellowship of the Ring, The (2001)           197           4.104
     858                                    Godfather, The (1972)           189           4.283
SUCCESS: 6. Popularity Model Training
Execution time: 2.80 seconds
```

STEP 7: 7. Collaborative Filtering Training

SUCCESS | Execution time: 7.70 seconds

```
============================================================
MODEL 2: ITEM-BASED COLLABORATIVE FILTERING
============================================================

Training interactions: 100,226
Users: 610
Movies: 9,702

Movie-user matrix shape: (9702, 610)

Collaborative filtering model trained.
Model saved to: C:\Users\Tanmay\Desktop\Recommendation-pipeline\models\collaborative_filtering_model.pkl

Collaborative filtering training completed successfully.

Top 5 personalized recommendations for User 1:

 movieId                                              title                 genres    score
    2918                          Ferris Bueller's Day Off (1986)                 Comedy 1.030889
     180                                          Mallrats (1995)         Comedy|Romance 0.994520
    4993 Lord of the Rings: The Fellowship of the Ring, The (2001)       Adventure|Fantasy 0.618892
    2762                                  Sixth Sense, The (1999)    Drama|Horror|Mystery 0.600661
    5418                               Bourne Identity, The (2002) Action|Mystery|Thriller 0.597148
SUCCESS: 7. Collaborative Filtering Training
Execution time: 7.70 seconds
```

STEP 8: 8. Model Evaluation

SUCCESS | Execution time: 26.76 seconds

```
============================================================
MODEL EVALUATION
============================================================

Users eligible for evaluation: 610
Training interactions: 100,226
Test interactions: 610
Training catalog size: 9,702

Evaluating popularity model...
Building collaborative filtering similarity matrix...
Evaluating collaborative filtering model...

============================================================
FINAL MODEL COMPARISON
============================================================
                 Model  K  Evaluated Users  Precision@5  Recall@5  NDCG@5  Coverage  Diversity
            Popularity  5              610       0.0052    0.0262  0.0145    0.0077     0.7790
Collaborative Filtering  5              610       0.0062    0.0311  0.0186    0.0612     0.7373

Results saved to: C:\Users\Tanmay\Desktop\Recommendation-pipeline\reports\model_evaluation.csv

Model evaluation completed successfully.
SUCCESS: 8. Model Evaluation
Execution time: 26.76 seconds
```

STEP 9: 9. MLflow Experiment Tracking

SUCCESS | Execution time: 15.35 seconds

```
2026/11/14 12:12:58 INFO mlflow.agent.hint: Load the `mlflow-trace-instrumentation` skill at
C:\Users\Tanmay\Desktop\Recommendation-pipeline\.venv\lib\site-packages\mlflow\assistant\skills\mlflow-trace-
instrumentation\SKILL.md before writing any tracing code; it ships with this MLflow install. Set
MLFLOW_DISABLE_AGENT_HINT=1 to silence this.
============================================================
MLFLOW EXPERIMENT TRACKING
============================================================

Experiment: MovieLens_Recommendation_System

Logged MLflow run: Popularity
 Precision@5: 0.0052
 Recall@5: 0.0262
 NDCG@5: 0.0145
 Coverage: 0.0077
 Diversity: 0.779

Logged MLflow run: Collaborative Filtering
 Precision@5: 0.0062
 Recall@5: 0.0311
 NDCG@5: 0.0186
 Coverage: 0.0612
 Diversity: 0.7373

MLflow experiment tracking completed successfully.
SUCCESS: 9. MLflow Experiment Tracking
Execution time: 15.35 seconds
```

PIPELINE EXECUTION EVIDENCE

Run completed: 2026-11-14 12:34:49 India Standard Time

Execution step screenshots: 9

The following pages show the captured terminal output and result for each executed step.

STEP 1: 1. Data Ingestion

SUCCESS | Execution time: 2.99 seconds

```
... 1 earlier output lines omitted ...
Data ingestion completed successfully!

Movies dataset:
Number of rows: 9742
Columns: ['movieId', 'title', 'genres']

Ratings dataset:
Number of rows: 100836
Columns: ['userId', 'movieId', 'rating', 'timestamp']

Sample movie records:
   movieId  ...                                       genres
0        1  ...  Adventure|Animation|Children|Comedy|Fantasy
1        2  ...                   Adventure|Children|Fantasy
2        3  ...                               Comedy|Romance
3        4  ...                         Comedy|Drama|Romance
4        5  ...                                       Comedy

[5 rows x 3 columns]

Sample rating records:
   userId  movieId  rating  timestamp
0       1        1     4.0  964982703
1       1        3     4.0  964981247
2       1        6     4.0  964982224
3       1       47     5.0  964983815
4       1       50     5.0  964982931
SUCCESS: 1. Data Ingestion
Execution time: 2.99 seconds
```

STEP 2: 2. Data Validation

SUCCESS | Execution time: 2.18 seconds

```
Starting data validation...

DATA VALIDATION RESULTS
=======================================
Missing values in movies:
movieId    0
title      0
genres     0
Missing values in ratings:
userId       0
movieId      0
rating       0
timestamp    0
Duplicate rows in movies: 0
Duplicate rows in ratings: 0
Movies columns validation: True
Ratings columns validation: True
Invalid movie field types: 0
Invalid rating field types: 0
Invalid ratings (outside 0.5-5.0 range or non-numeric): 0
Invalid movie references in ratings: 0
Invalid timestamps in ratings: 0

Data validation completed successfully!
Validation report saved to: C:\Users\Tanmay\Desktop\Recommendation-pipeline\data\errors\validation_report.txt
SUCCESS: 2. Data Validation
Execution time: 2.18 seconds
```

STEP 3: 3. Data Preparation

SUCCESS | Execution time: 4.04 seconds

```
... 7 earlier output lines omitted ...
Duplicate removal:
Movies duplicates removed : 0
Ratings duplicates removed: 0

============================================================
PREPARATION RESULTS
============================================================

Clean movies           : 9,742
Clean interactions     : 100,836
Training interactions  : 100,226
Testing interactions   : 610
Unique users           : 610
Unique movies rated    : 9,724
Minimum rating         : 0.5
Maximum rating         : 5.0
First interaction      : 1996-03-29 18:36:55
Last interaction       : 2018-09-24 14:27:30

Files created:
data/processed/movies_clean.csv
data/processed/ratings_clean.csv
data/processed/train.csv
data/processed/test.csv
data/processed/users.csv

Data preparation completed successfully.
SUCCESS: 3. Data Preparation
Execution time: 4.04 seconds
```

STEP 4: 4. Data Transformation

SUCCESS | Execution time: 14.06 seconds

```
... 8 earlier output lines omitted ...
3       4               216           216          3.556 2001-12-05 17:49:02
4       5                44            44          3.636 1996-11-08 06:42:17

Most popular movies:
      movieId  ... average_rating
314       356  ...          4.164
277       318  ...          4.429
257       296  ...          4.197
510       593  ...          4.161
1938     2571  ...          4.192
224       260  ...          4.231
418       480  ...          3.750
97        110  ...          4.032
507       589  ...          3.971
461       527  ...          4.225

[10 rows x 4 columns]

User-item matrix shape:
(610, 9724)

Files created:
data/features/user_activity.csv
data/features/item_popularity.csv
data/features/user_item_matrix.csv

Transformation completed successfully.
SUCCESS: 4. Data Transformation
Execution time: 14.06 seconds
```

STEP 5: 5. Feature Engineering

SUCCESS | Execution time: 4.36 seconds

```
... 23 earlier output lines omitted ...
Feature Catalog:
                      feature  ...                                        description
0                      userId  ...  MovieLens user identifier; no profile attribut...
1              user_avg_rating  ...                   Average rating given by the user
2          user_activity_count  ...                Total number of user interactions
3           user_unique_movies  ...             Number of unique movies rated by user
4   days_since_last_interaction  ...         Days since user's most recent interaction
5                      movieId  ...                           MovieLens item identifier
6              item_popularity  ...                  Total number of ratings received
7            item_unique_users  ...           Number of unique users rating the movie
8              item_avg_rating  ...                  Average rating received by movie
9                       genres  ...                           Movie genre information
10                      rating  ...    Explicit rating supplied by a user for a movie
11           interaction_time  ...       UTC timestamp converted to a datetime value
12           interaction_hour  ...         Hour of day when the interaction occurred
13     interaction_day_of_week  ...  Day of week when the interaction occurred, Mon...
14           interaction_month  ...                Month when the interaction occurred

[15 rows x 4 columns]

Files created:
data/features/user_features.csv
data/features/item_features.csv
data/features/interaction_features.csv
reports/feature_catalog.csv

Feature engineering completed successfully.
SUCCESS: 5. Feature Engineering
Execution time: 4.36 seconds
```

STEP 6: 6. Popularity Model Training

SUCCESS | Execution time: 2.83 seconds

```
... 6 earlier output lines omitted ...
```

movieId	title	rating_count	average_rating
356	Forrest Gump (1994)	329	4.164
318	Shawshank Redemption, The (1994)	314	4.430
296	Pulp Fiction (1994)	305	4.192
2571	Matrix, The (1999)	278	4.192
593	Silence of the Lambs, The (1991)	278	4.162
260	Star Wars: Episode IV - A New Hope (1977)	251	4.231
480	Jurassic Park (1993)	236	3.748
110	Braveheart (1995)	235	4.032
589	Terminator 2: Judgment Day (1991)	221	3.959
2959	Fight Club (1999)	218	4.273

```
Movies eligible for recommendation: 3,633

Model saved to: C:\Users\Tanmay\Desktop\Recommendation-pipeline\models\popularity_model.pkl

Popularity model training completed successfully.

Top 5 recommendations for User 1:
```

movieId	title	rating_count	average_rating
318	Shawshank Redemption, The (1994)	314	4.430
589	Terminator 2: Judgment Day (1991)	221	3.959
150	Apollo 13 (1995)	200	3.840
4993	Lord of the Rings: The Fellowship of the Ring, The (2001)	197	4.104
858	Godfather, The (1972)	189	4.283

```
SUCCESS: 6. Popularity Model Training
Execution time: 2.83 seconds
```

STEP 7: 7. Collaborative Filtering Training

SUCCESS | Execution time: 7.16 seconds

```
============================================================
MODEL 2: ITEM-BASED COLLABORATIVE FILTERING
============================================================

Training interactions: 100,226
Users: 610
Movies: 9,702

Movie-user matrix shape: (9702, 610)

Collaborative filtering model trained.
Model saved to: C:\Users\Tanmay\Desktop\Recommendation-pipeline\models\collaborative_filtering_model.pkl

Collaborative filtering training completed successfully.

Top 5 personalized recommendations for User 1:

 movieId                                              title                 genres    score
    2918                          Ferris Bueller's Day Off (1986)                 Comedy 1.030889
     180                                          Mallrats (1995)         Comedy|Romance 0.994520
    4993 Lord of the Rings: The Fellowship of the Ring, The (2001)       Adventure|Fantasy 0.618892
    2762                                  Sixth Sense, The (1999)    Drama|Horror|Mystery 0.600661
    5418                               Bourne Identity, The (2002) Action|Mystery|Thriller 0.597148
SUCCESS: 7. Collaborative Filtering Training
Execution time: 7.16 seconds
```

STEP 8: 8. Model Evaluation

SUCCESS | Execution time: 25.12 seconds

```
================================================================
MODEL EVALUATION
================================================================

Users eligible for evaluation: 610
Training interactions: 100,226
Test interactions: 610
Training catalog size: 9,702

Evaluating popularity model...
Building collaborative filtering similarity matrix...
Evaluating collaborative filtering model...

================================================================
FINAL MODEL COMPARISON
================================================================
                 Model  K  Evaluated Users  Precision@5  Recall@5  NDCG@5  Coverage  Diversity
            Popularity  5              610       0.0052    0.0262  0.0145    0.0077     0.7790
Collaborative Filtering  5              610       0.0062    0.0311  0.0186    0.0612     0.7373

Results saved to: C:\Users\Tanmay\Desktop\Recommendation-pipeline\reports\model_evaluation.csv

Model evaluation completed successfully.
SUCCESS: 8. Model Evaluation
Execution time: 25.12 seconds
```

STEP 9: 9. MLflow Experiment Tracking

SUCCESS | Execution time: 18.21 seconds

```
============================================================
MLFLOW EXPERIMENT TRACKING
============================================================

Experiment: MovieLens_Recommendation_System

Logged MLflow run: Popularity
 Precision@5: 0.0052
 Recall@5: 0.0262
 NDCG@5: 0.0145
 Coverage: 0.0077
 Diversity: 0.779

Logged MLflow run: Collaborative Filtering
 Precision@5: 0.0062
 Recall@5: 0.0311
 NDCG@5: 0.0186
 Coverage: 0.0612
 Diversity: 0.7373

MLflow experiment tracking completed successfully.
SUCCESS: 9. MLflow Experiment Tracking
Execution time: 18.21 seconds
```